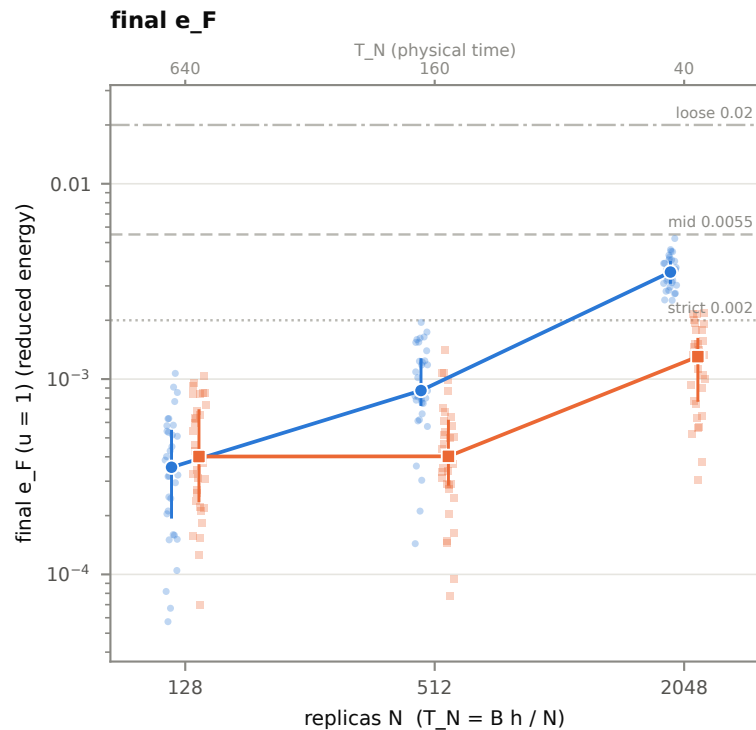
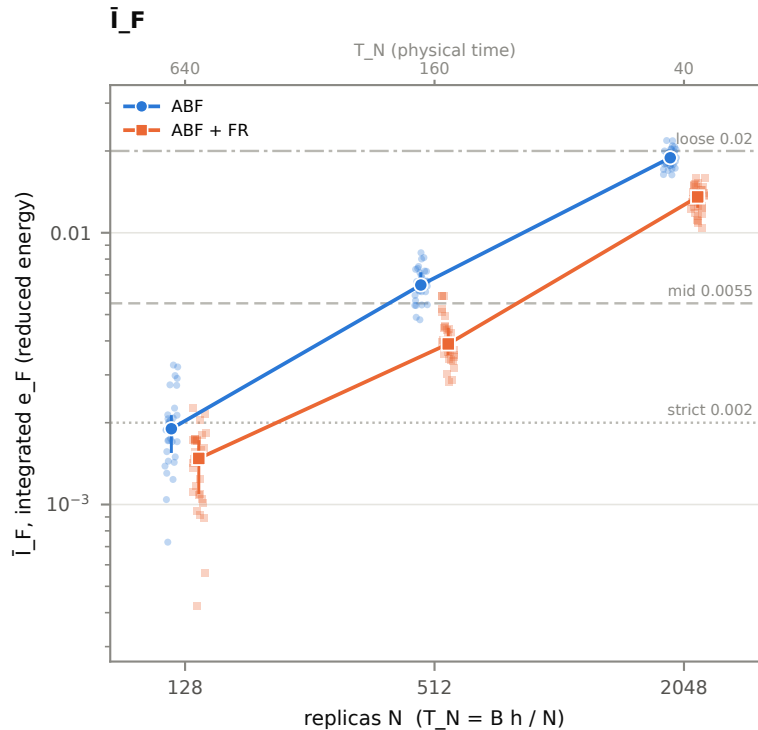


gateway alpha0.5: absolute free-energy error vs N

Equal total budget per arm per seed: $B = N \times n_steps = 3.2768e+09$ walker-steps at every N ($h = 2.5e-05$, $T_N = B \cdot h / N$); 32 seeds per (N, method). ABF and ABF + FR: seed median, IQR and per-seed points.



gateway alpha0.5 · S1_free_energy_abs_vs_N · status: all planned (N, seed, method) complete · FR is not defined at N = 1 (ABF only, by design).
Markers: seed median; whiskers: IQR over seeds; faint points: individual seeds. Grey lines: the frozen strict / mid / loose thresholds of $e(t)$ (shown for scale on the integrated errors as well).
Reference floors: lbar_F: EM bias of F vs analytic $2.8e-06$ lies below the axis; final_e_F: EM bias of F vs analytic $2.8e-06$ lies below the axis.
Source results/mechanism/matched_free_energy/alpha0.5/analysis/summary.json (generated 2026-10-10T14:19:19Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.